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(54) **SEMICONDUCTOR PACKAGE INCLUDING
ELECTROMAGNETIC SHIELD STRUCTURE**

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(2013.01); **H01L 2924/3025** (2013.01)

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H01L 24/73; **H01L 24/81**; **H01L 25/105**

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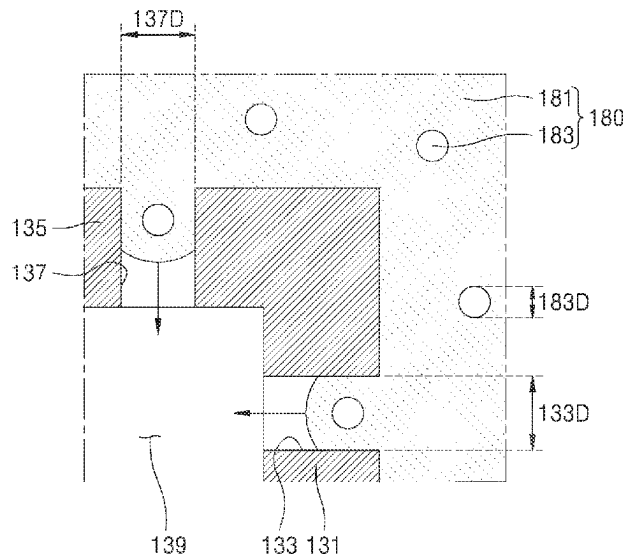
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(57) **ABSTRACT**

A semiconductor package includes; a package substrate, a semiconductor chip on the package substrate, an electromagnetic shield structure on the package substrate and including an upper cover covering an upper surface of the semiconductor chip and a side cover surrounding the semiconductor chip, and a sealing member contacting the semiconductor chip and the electromagnetic shield structure, wherein the side cover includes first through holes and the upper cover includes second through holes.

20 Claims, 11 Drawing Sheets



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FIG. 1

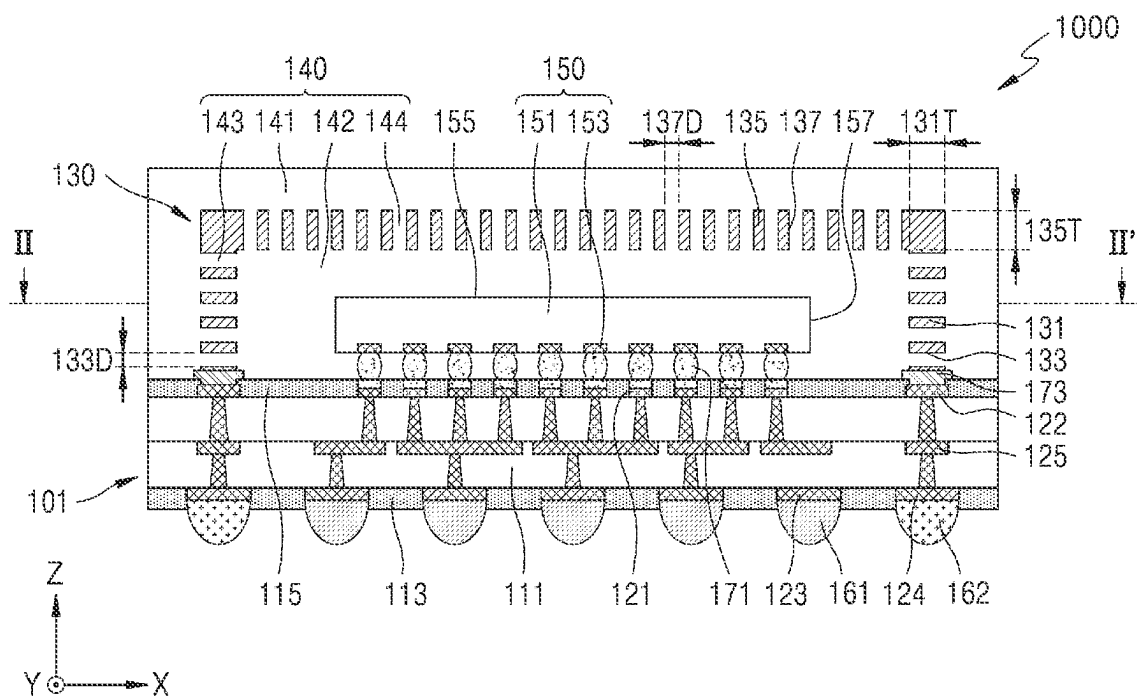


FIG. 2

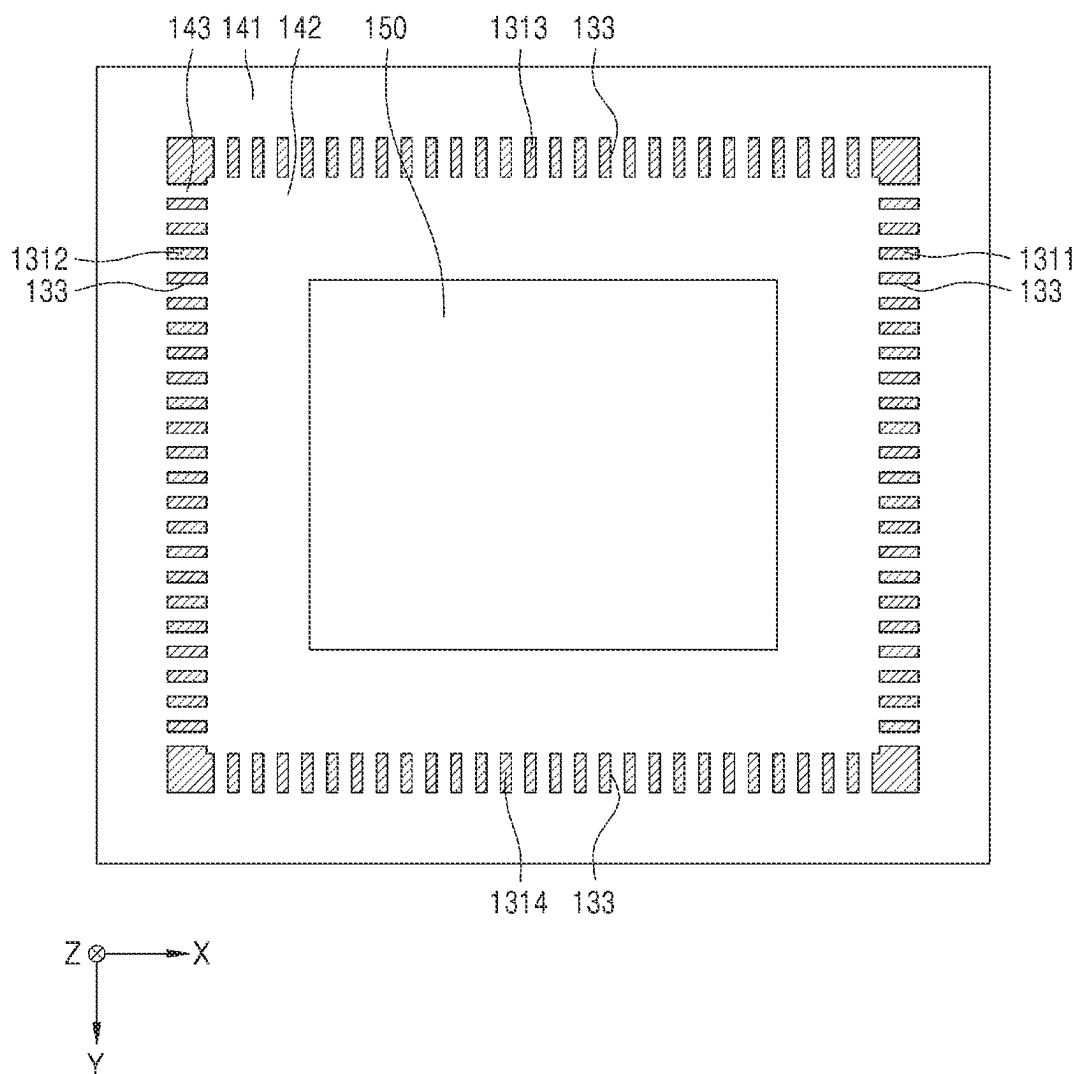


FIG. 3

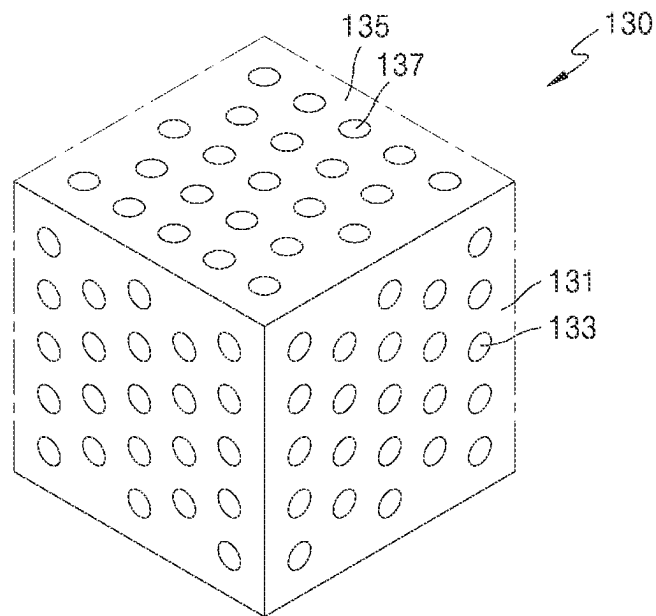


FIG. 4

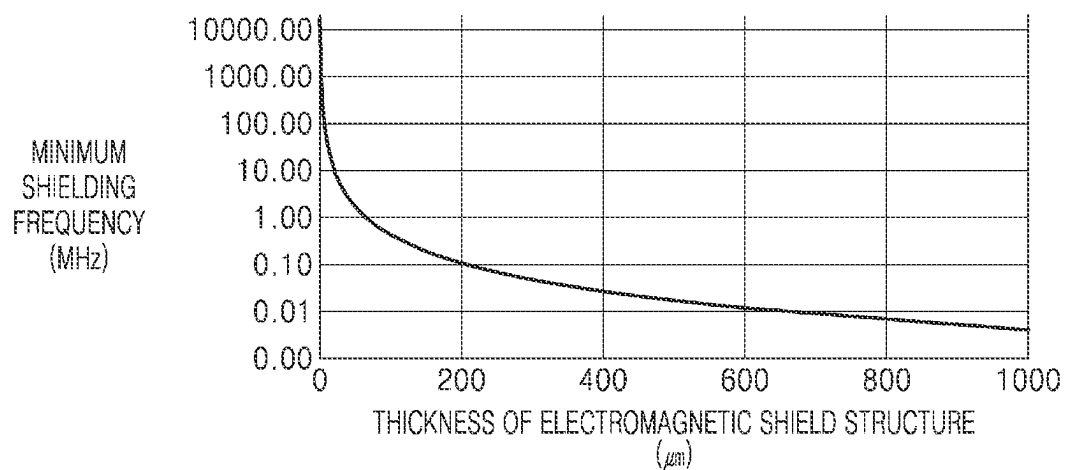


FIG. 5

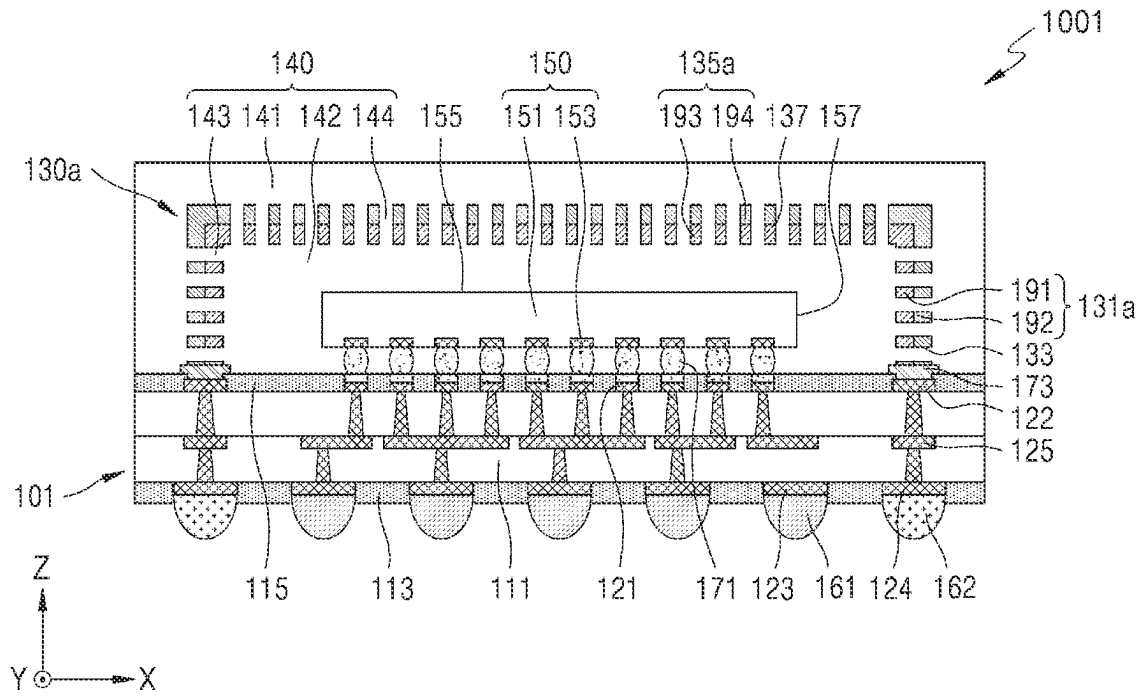


FIG. 6

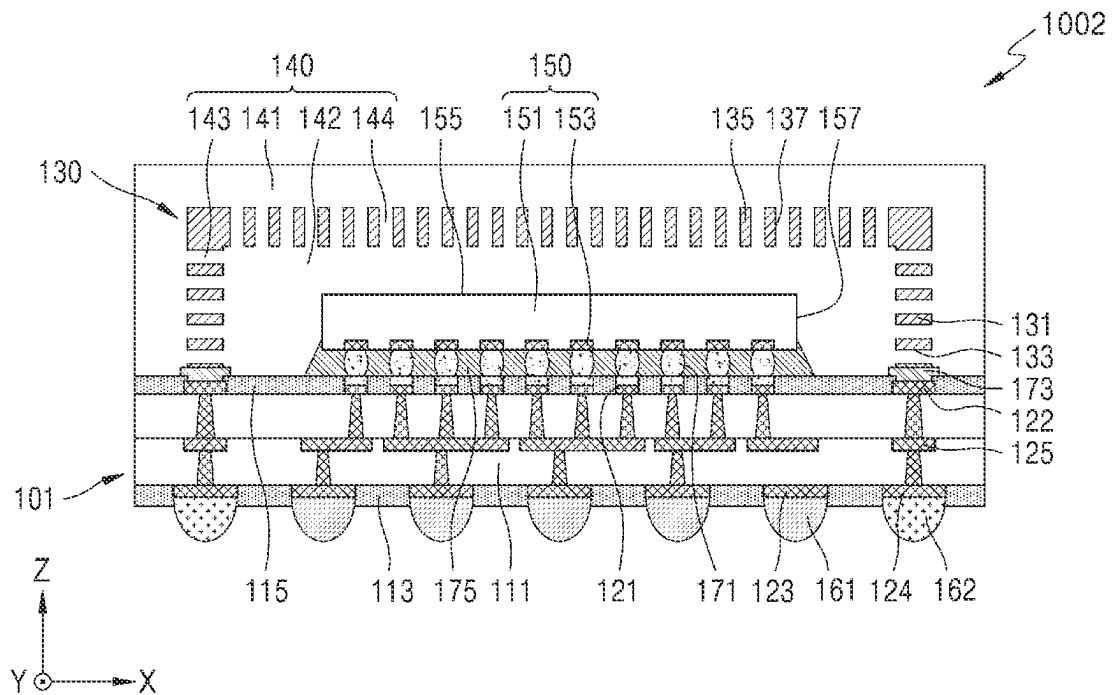


FIG. 7

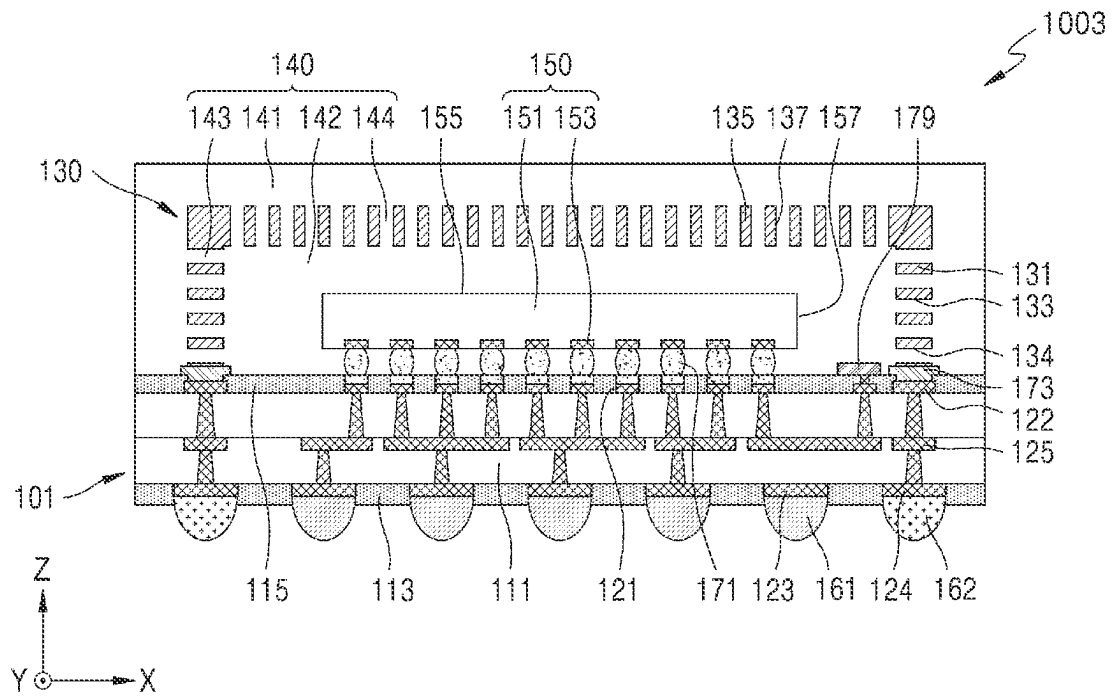


FIG. 8A

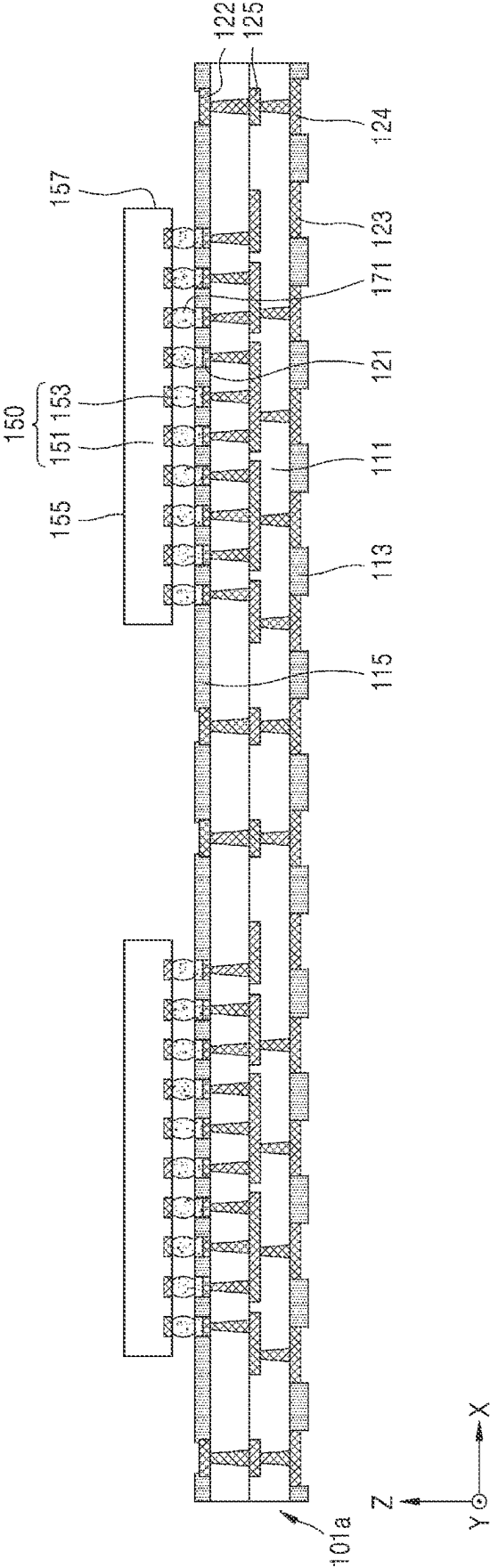


FIG. 8B

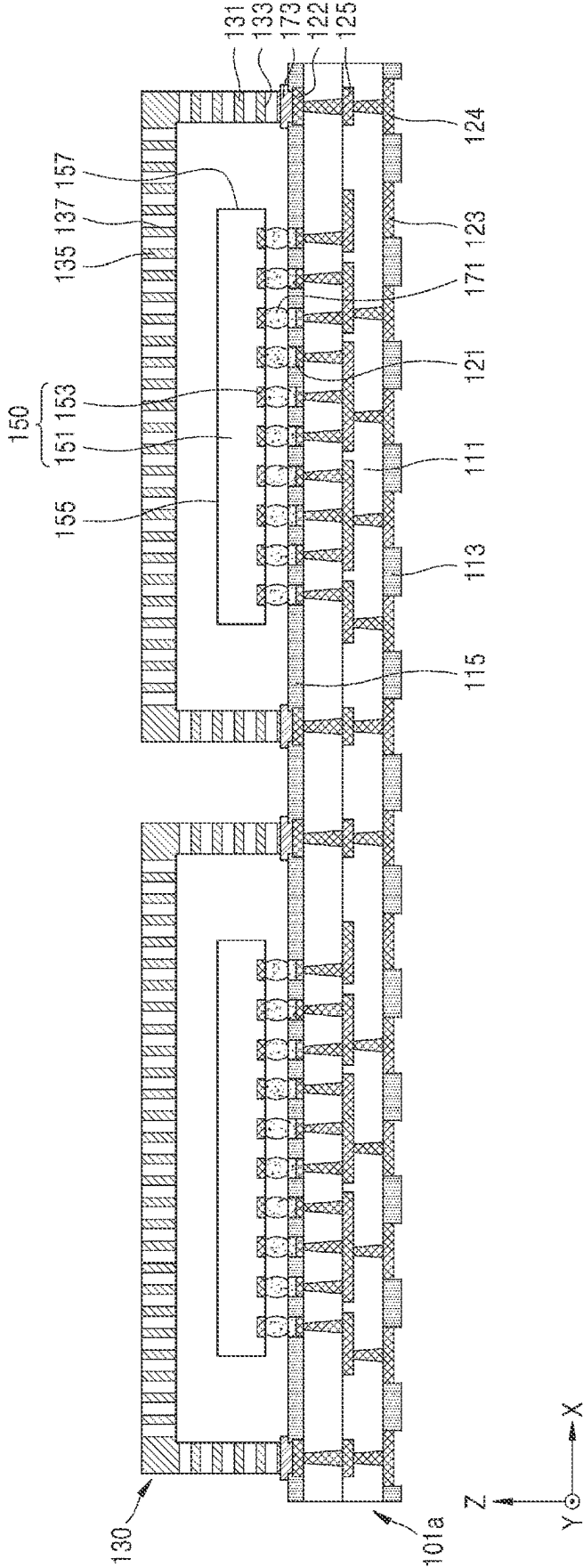


FIG. 8C

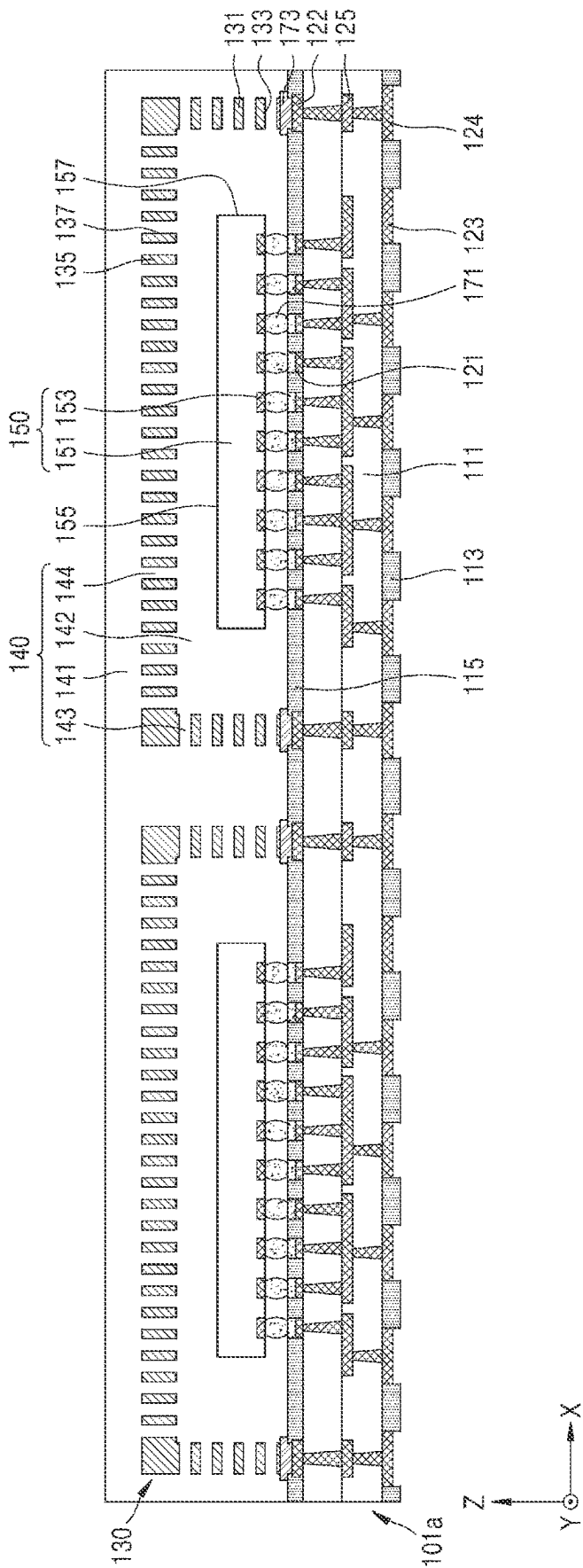


FIG. 8D

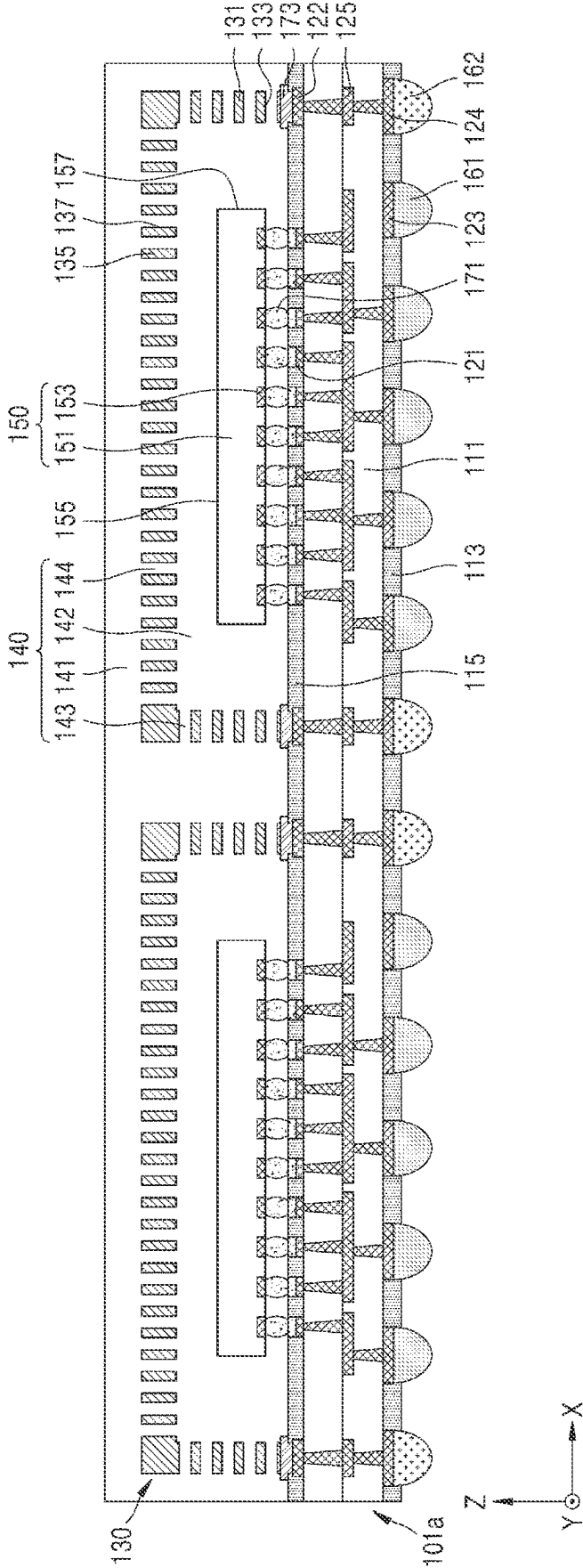


FIG. 8E

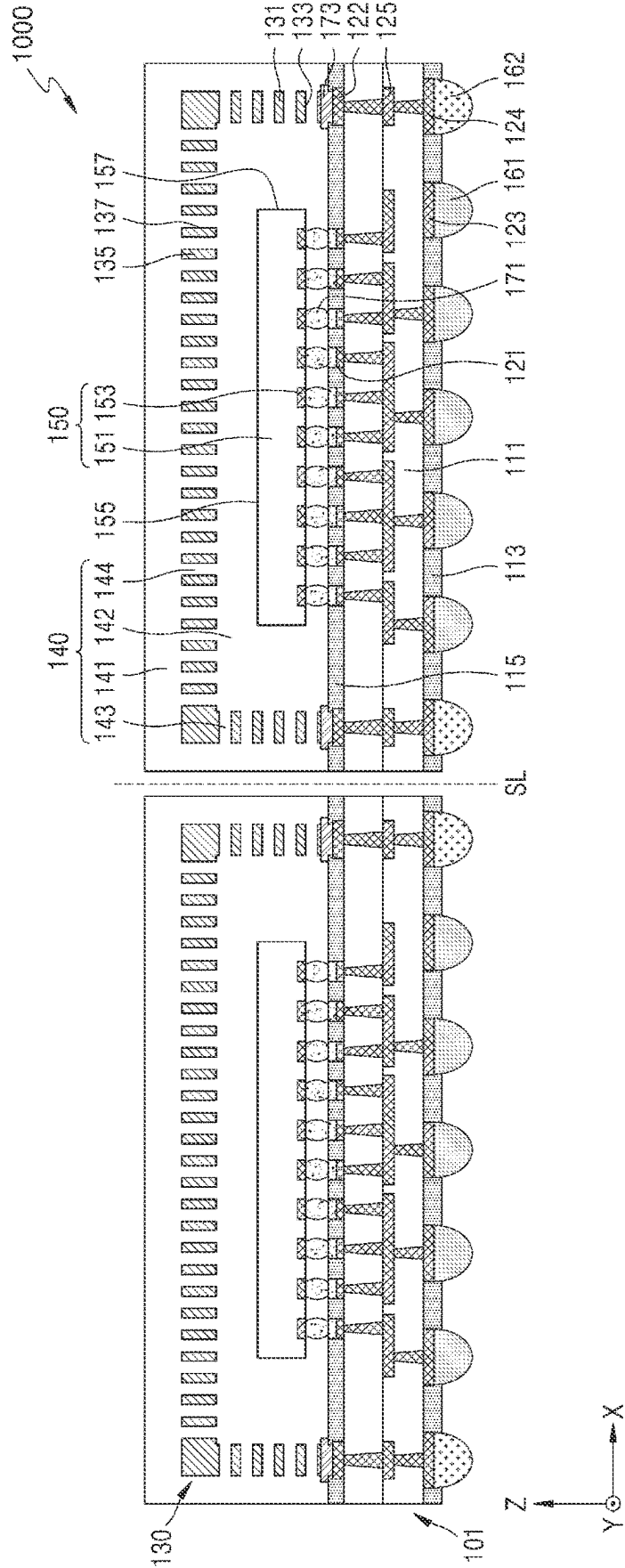
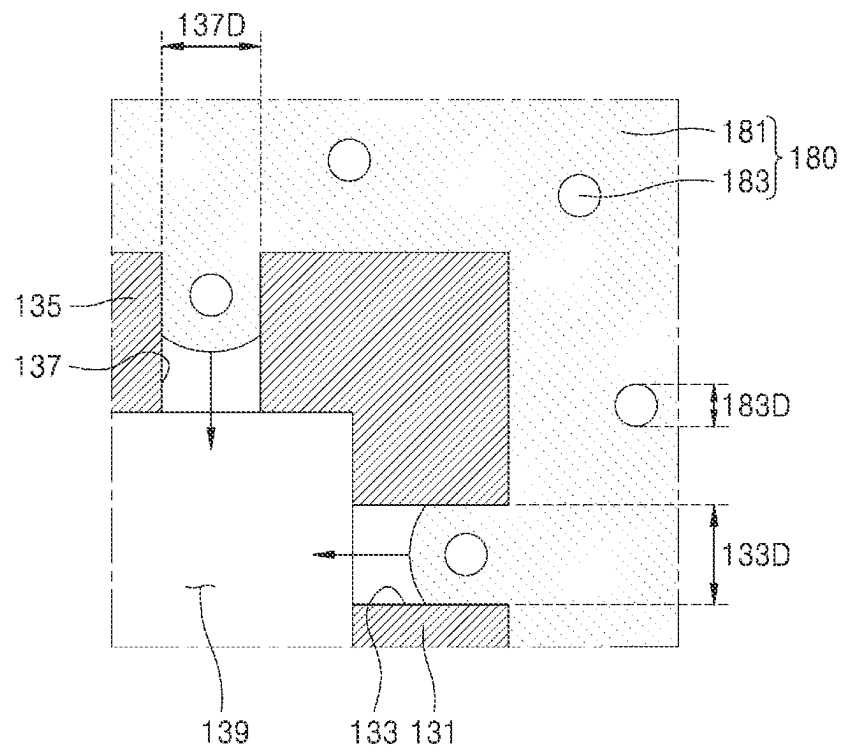


FIG. 9



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SEMICONDUCTOR PACKAGE INCLUDING ELECTROMAGNETIC SHIELD STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0135217 filed on Oct. 12, 2021 in the Korean Intellectual Property Office, the subject matter of which is hereby incorporated by reference in its entirety.

BACKGROUND

The inventive concept relates generally to semiconductor packages, and more particularly to semiconductor packages including an electromagnetic shield structure.

Increasing integration density for electronic devices caused by continuing reductions in physical size, together with consumer demands for expanded functionality and improved performance place enormous demands on constituent electronic components, such as semiconductor devices. For example, improved shielding against electromagnetic interference (EMI) is an increasingly noteworthy consideration in the design and fabrication of electronic components and electronic devices including same.

SUMMARY

Embodiments of the inventive concept provide semiconductor devices including an electromagnetic shield structure, wherein such embodiments of the inventive concept exhibit improved immunity to electromagnetic interference.

According to an aspect of the inventive concept, a semiconductor package may include; a package substrate, a semiconductor chip on the package substrate, an electromagnetic shield structure on the package substrate and including an upper cover covering an upper surface of the semiconductor chip and a side cover surrounding the semiconductor chip, and a sealing member contacting the semiconductor chip and the electromagnetic shield structure, wherein the side cover includes first through holes and the upper cover includes second through holes.

According to an aspect of the inventive concept, a semiconductor package may include; a package substrate including a base, first upper substrate pads on an upper surface of the package substrate, second upper substrate pads on the upper surface of the package substrate, and an internal interconnection pattern within the base and connected to the first upper substrate pads, a semiconductor chip including chip connecting bumps respectively connected to the first upper substrate pads, an electromagnetic shield structure on the package substrate and including an upper cover covering an upper surface of the semiconductor chip and a side cover surrounding the semiconductor chip, wherein the side cover is electrically connected to the second upper substrate pads and ground, and a sealing member between the semiconductor chip and the electromagnetic shield structure, wherein the side cover includes first through holes, and the sealing member fills the first through holes.

According to an aspect of the inventive concept, a semiconductor package may include; a package substrate including first upper substrate pads on an upper surface of the package substrate and second upper substrate pads connected to ground and on the upper surface of the package substrate, a radio frequency integrated circuit mounted on the package substrate using chip connecting bumps respec-

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tively connected to the first upper substrate pads, wherein the radio frequency integrated circuit is configured to communicate a radio frequency signal having a wavelength λ , an electromagnetic shield structure mounted on the package substrate using the second upper substrate pads and including an upper cover covering an upper surface of the radio frequency integrated circuit and a side cover surrounding side surfaces of the radio frequency integrated circuit, wherein the side cover is penetrated by regularly spaced first through holes and the upper cover is penetrated by regularly spaced second through holes, and a sealing member disposed between the electromagnetic shield structure and the radio frequency integrated circuit, surrounding the chip connecting bumps, and including a filler having a particle size, wherein a thickness of the upper cover and a thickness of the side cover ranges between about 200 μm to about 1,000 μm , a first size of the first through holes and a second size of the second through holes are each at least twice that of the particle size, and the first size of the first through holes and the second size of the second through holes are each less than or equal to $\lambda/50$.

BRIEF DESCRIPTION OF THE DRAWINGS

Advantages, benefits, features and related aspects, as well as the making and use of the inventive concept may be more clearly understood upon consideration of the following detailed description together with the accompanying drawings, in which:

FIG. 1 is a cross-sectional view illustrating a semiconductor package according to embodiments of the inventive concept;

FIG. 2 is a plan (or top-down) view taken along line II-II' of FIG. 1 and further illustrating the semiconductor package of FIG. 1;

FIG. 3 is a perspective view illustrating, in part, an electromagnetic shield structure associated with semiconductor packages according to embodiments of the inventive concept;

FIG. 4 is a graph showing a relationship between a thickness of an electromagnetic shield structure and the minimum shielding frequency shielded by the electromagnetic shield structure;

FIGS. 5, 6 and 7 are respective cross-sectional views illustrating various semiconductor packages according to embodiments of the inventive concept;

FIGS. 8A, 8B, 8C, 8D and 8E (hereafter collectively, "FIGS. 8A to 8E") are related cross-sectional views illustrating in one example a method of manufacturing a semiconductor package according to embodiments of the inventive concept; and

FIG. 9 is a cross-sectional view of further illustrating the forming of the sealing member in relation to the method of FIGS. 8A to 8E.

DETAILED DESCRIPTION

Throughout the written description and drawings, like reference numbers and labels are used to denote like or similar elements, components, method steps and/or features. Throughout the written description certain geometric terms may be used to highlight relative relationships between elements, components and/or features with respect to certain embodiments of the inventive concept. Those skilled in the art will recognize that such geometric terms are relative in nature, arbitrary in descriptive relationship(s) and/or directed to aspect(s) of the illustrated embodiments. Geo-

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metric terms may include, for example: height/width; vertical/horizontal; top/bottom; higher/lower; closer/farther; thicker/thinner; proximate/distant; above/below; under/over; upper/lower; center/side; surrounding; overlay/underlay; etc.

FIG. 1 is a cross-sectional view illustrating a semiconductor package 1000 according to embodiments of the inventive concept, FIG. 2 is a plan view taken along line II-II' of the semiconductor package 1000, and FIG. 3 is a perspective view further illustrating, in part, an electromagnetic shield structure 130 of the semiconductor package 1000.

Referring to FIGS. 1, 2 and 3, the semiconductor package 1000 may include a package substrate 101, a semiconductor chip 150, an electromagnetic shield structure 130, and a sealing member 140.

The package substrate 101 may include a substrate base 111, first upper substrate pads 121, second upper substrate pads 122, first lower substrate pads 123, second lower substrate pads 124, and an internal interconnect pattern 125. The package substrate 101 may be, for example, a printed circuit board (PCB).

The substrate base 111 may include at least one of phenol resin, epoxy resin, and polyimide.

The first upper substrate pads 121 and the second upper substrate pads 122 may be provided on an upper surface of the substrate base 111. The first lower substrate pads 123 and the second lower substrate pads 124 may be provided on a lower surface of the substrate base 111.

The internal interconnect pattern 125 may be variously configured and provided within, wholly or in part, the substrate base 111. Thus, the internal interconnect pattern 125 may variously and electrically connect the first upper substrate pads 121, the second upper substrate pads 122, the first lower substrate pads 123, and/or the second lower substrate pads 124. In some embodiments, the internal interconnect pattern 125 may include line patterns extending in at least one of a first (or X-) horizontal direction and a second (or Y-) horizontal direction within the substrate base 111. Alternately or additionally, the internal interconnection pattern 125 may include via patterns extending in a vertical (or Z-) direction within the substrate base 111.

The package substrate 101 may further include an upper protective layer 115 provided on the upper surface of the substrate base 111 and a lower protective layer 113 provided on the lower surface of the substrate base 111. Here, the upper protective layer 115 and the lower protective layer 113 may include, for example, solder resist.

The first upper substrate pads 121, the second upper substrate pads 122, the first lower substrate pads 123, the second lower substrate pads 124, and/or the internal interconnect pattern 125 may variously include at least one of, for example, copper (Cu), aluminum (Al), tungsten (W), titanium (Ti), tantalum (Ta), indium (In), molybdenum (Mo), manganese (Mn), cobalt (Co), tin (Sn), nickel (Ni), magnesium (Mg), rhenium (Re), beryllium (Be), gallium (Ga), and ruthenium (Ru).

The first upper substrate pads 121 may be used to respectively connect chip connecting bumps 171 (e.g., micro bumps) associated with the semiconductor chip 150 to the semiconductor chip 150. That is, the chip connecting bumps 171 may be respectively connected to the first upper substrate pads 121 through openings provided in the upper protective layer 115.

The second upper substrate pads 122 may be used to connect (or attach) the electromagnetic shield structure 130. In this regard, a conductive bonding material layer 173 may

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be used to electrically connect the second upper substrate pads 122 and the electromagnetic shield structure 130. Hence, the conductive bonding material layer 173 may be provided between the second upper substrate pads 122 and the electromagnetic shield structure 130. Further in this regard, the conductive bonding material layer 173 may be connected to the second upper substrate pads 122 through openings provided in the upper protective layer 115. In some embodiments, the conductive bonding material layer 173 may include solder.

The first lower substrate pads 123 may be used to connect (or attach) first external connection terminals 161 and second external connection terminals 162.

The first external connection terminals 161 may be connected to the first lower substrate pads 123 through openings provided in the lower protective layer 113. Accordingly, the first external connection terminals 161 may be respectively and electrically connected to the first upper substrate pads 121 through the internal interconnect pattern 125. In this manner, the first external connection terminals 161 may be configured to electrically and physically connect the semiconductor package 1000 with an external device, and may be variously used as terminal(s) for communicating (e.g., transmitting and/or receiving) input/output (I/O) signals (e.g., at least one of power signals, a ground signal, command signals, control signals and/or address signals). In some embodiments, the first external connection terminals 161 may be configured to receive at least one power signal from an external device.

The external connection terminals 162 may be connected to the second lower substrate pads 124 through openings in the lower protective layer 113, and may be electrically connected through the second lower substrate pads 124 to the internal interconnect pattern 125. In some embodiments, the second external connection terminals 162 may be configured to receive a ground signal from an external device.

In some embodiments, the first external connection terminals 161 and/or the second external connection terminals 162 may be formed from, for example, solder balls or solder bumps.

The semiconductor chip 150 mounted on (e.g., mechanically assembled and/or electrically connected) the package substrate 101 may include a semiconductor substrate 151 and chip pads 153, wherein the chip pads 153 may be provided on a lower surface of the semiconductor chip 150.

In some embodiments, the semiconductor chip 150 may be mounted on the package substrate 101 in a face-down (or flip-chip) manner. That is, the semiconductor chip 150 may be mounted on the package substrate 101, such that the lower surface of the semiconductor chip 150 faces the package substrate 101. The chip pads 153 of the semiconductor chip 150 may be electrically connected to the first upper substrate pads 121 through the chip connecting bumps 171. The chip pads 153 of the semiconductor chip 150 may be used as terminals for variously communicating I/O signals associated with the semiconductor chip 150.

The semiconductor substrate 151 may include silicon (e.g., silicon, polycrystalline silicon, or amorphous silicon). The semiconductor substrate 151 may include an active surface and an opposing inactive surface, wherein the active surface of the semiconductor substrate 151 is the lower surface of the semiconductor chip 150, and the inactive surface of the semiconductor substrate 151 is the an upper surface 155 of the semiconductor chip 150. The semiconductor chip 150 may include a semiconductor device layer formed on the active surface of the semiconductor substrate 151. The chip pads 153 of the semiconductor chip 150 may

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be used to electrically connect to the semiconductor device layer through a wiring structure (not shown) provided (e.g.) within the semiconductor chip **150**.

In some embodiments, the semiconductor chip **150** may include a volatile memory chip and/or a non-volatile memory chip. Here, the volatile memory chip may include, for example, a dynamic random access memory (RAM) (DRAM), a static RAM (SRAM), a thyristor RAM (TRAM), a zero capacitor RAM (ZRAM), or a twin transistor RAM (TTRAM). The non-volatile memory chip may include, for example, a flash memory, a magnetic RAM (MRAM), a spin-transfer torque MRAM (STT-MRAM), a ferroelectric RAM (FRAM), a phase change RAM (PRAM), a resistive RAM (RRAM), a nanotube RRAM (RRAM), a polymer RAM (polymer RAM), or an insulator resistance change memory.

Alternately or additionally, the semiconductor chip **150** may be a logic chip. The logic chip may include, for example, an artificial intelligence semiconductor, a micro-processor, a graphics processor, a signal processor, a network processor, a chipset, an audio codec, a video codec, and an application processor.

In some embodiments, the semiconductor chip **150** is a communication chip and may include a signal processing circuit for processing radio frequency (RF) signal(s). That is, in some embodiments, the semiconductor chip **150** may include a radio-frequency integrated circuit (RFIC).

Hence, in various embodiments, the semiconductor package **1000** may include one or more semiconductor chip(s) **150**. For example, two or more, spaced apart semiconductor chips **150** may be arranged on the upper surface of the package substrate **101**, wherein the two or more semiconductor chips **150** may be the same chip type, or different chip types.

The electromagnetic shield structure **130** may be mounted on the package substrate **101** to be vertically (e.g., in the vertical direction) and laterally (e.g., in the first and second horizontal directions) spaced apart from the semiconductor chip **150**. The electromagnetic shield structure **130** may be mounted on the package substrate **101** using a surface mounting technique, for example. In this regard, the electromagnetic shield structure **130** may be physically and electrically connected to the second upper substrate pads **122** of the package substrate **101** through the conductive bonding material layer **173**.

In its configuration, the electromagnetic shield structure **130** may substantially cover (or encompass) an upper surface **155** and side surfaces **157** of the semiconductor chip **150**. For example, the electromagnetic shield structure **130** may have a cup-like shape that covers the semiconductor chip **150** and defines an internal space accommodating the semiconductor chip **150**. (See, e.g., element **139** of FIG. 9). Hence, in various embodiments, the electromagnetic shield structure **130** may substantially cover the upper surface **155** of the semiconductor chip **150** and substantially surround the side surfaces **157** of the semiconductor chip **150**. In some embodiments, a thickness of the electromagnetic shield structure **130** may be substantially uniform.

It its shape and constituent nature, the electromagnetic shield structure **130** will “shield” (or suppress) the emission of electromagnetic energy generated by the semiconductor chip **150** during its operation, such that the potentially interfering electromagnetic energy (e.g., electromagnetic waves, electrical frequencies and/or magnetic fields) is not communicated the outside the semiconductor package **1000**. Additionally, the electromagnetic shield structure **130** will shield the semiconductor chip **150** from potentially interfer-

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ing electromagnetic energy generated outside the semiconductor package **1000**. For example, the electromagnetic shield structure **130** may be configured to shield electromagnetic interference by reflecting and/or absorbing electromagnetic energy. In some embodiments, the electromagnetic shield structure **130** may be configured to shield electromagnetic energy in one or more defined frequency band(s).

In this regard, the electromagnetic shield structure **130** may include at least one of, for example, metal, magnetic material, carbon, ceramic, polymer, etc. In some embodiments, the electromagnetic shield structure **130** may include Cu, silver (Ag), and/or ferrite. The electromagnetic shield structure **130** may have a single-layer structure or a multi-layer structure. For example, the multi-layer electromagnetic shield structure **130** may include a second material layer stacked on a first material layer, different from the first material layer.

Accordingly, in some embodiments, the second upper substrate pads **122** may be electrically grounded. Hence, the second upper substrate pads **122** may be referred to as “ground pads.” For example, the second upper substrate pads **122** may be connected to the second external connection terminals **162**, which are connected to ground through the internal interconnect pattern **125**. The conductive bonding material layer **173**, the second upper substrate pads **122**, the internal interconnect pattern **125**, and the second lower substrate pads **124** may form an electrical connection path between the electromagnetic shield structure **130** and the second external connection terminal **162** through which electromagnetic energy may be absorbed (or transferred) away from the semiconductor chip **150** using the electromagnetic shield structure **130**.

In some embodiments, the electromagnetic shield structure **130** may be understood as including a side cover **131** laterally surrounding the semiconductor chip **150** and an upper cover **135** covering the upper surface **155** of the semiconductor chip **150**.

The upper cover **135** of the electromagnetic shield structure **130** may have a flat plate-like shape (e.g., a rectangular, flat, plate-like shape). The planar (or lateral) area of the upper cover **135** may be greater than a planar area of the semiconductor chip **150**, and the upper cover **135** may entirely cover the upper surface **155** of the semiconductor chip **150**.

The side cover **131** of the electromagnetic shield structure **130** may have a ring-like shape continuously extending along the circumference of the semiconductor chip **150** to surround the semiconductor chip **150**. For example, the side cover **131** may have a rectangular, ring-like shape. Thus, the side cover **131** may be spaced apart from, and respectively face (e.g., laterally oppose in one of the first horizontal direction and the second horizontal direction) the sidewalls **157** of the semiconductor chip **150**. A height of the side cover **131** may be greater than a height of the side surfaces **157** of the semiconductor chip **150**.

The side cover **131** may extend from outer edges of the upper cover **135** to the upper surface of the package substrate **101**. In some embodiments, the side cover **131** may extend substantially linearly (e.g., vertically) from the outer edges of the upper cover **135** to the upper surface of the package substrate **101**, wherein a lower edge of the side cover **131** may contact the conductive bonding material layer **173**.

The side cover **131** may include a first through hole **133**. The first through hole **133** may laterally penetrate (e.g., in the X-direction and/or the Y-direction) through the side

cover **131**, wherein the first through holes **133** may be spaced apart one from the other in the side cover **131**. In some embodiments, the first through holes **133** may be regularly or uniformly spaced in the side cover **131** according to a defined density. Here, the “density” of the first through holes **133** may be understood as a number of first through holes **133** in a unit area of the side cover **131**. In some embodiments, arrangement(s) of the first through holes **133** may be spaced apart at regular intervals. In some embodiments, the first through holes **133** may have the same dimension(s) (e.g., size, diameter, shape, etc.).

The upper cover **135** may include second through holes **137**, wherein the second through holes **137** may vertically penetrate the upper cover **135**. The second through holes **137** may be regularly spaced apart from one another in the upper cover **135** according to a defined density. In some embodiments, arrangement(s) of the second through holes **137** may be regularly spaced apart. In some embodiments, the second through holes **137** may have the same dimension(s) (e.g., size, diameter, shape, etc.).

In some embodiments, a first size **133D** for the first through hole **133** and a second size **137D** for the second through holes **137** may be the same. Here, the first size **133D** may denote a maximum width (or diameter) for the first through holes **133**, and the second size **137D** may denote a maximum width (or diameter) for the second through holes **137**.

In some embodiments, when a distance between respective centers of two, adjacent first through holes **133** may define a first pitch, and a distance between respective centers of two, adjacent second through holes **137** may define a second pitch. In some embodiments, the first pitch may be the same as the second pitch.

The sealing member **140** may be provided on the package substrate **101**. The sealing member **140** may cover at least a portion of the package substrate **101**, at least a portion of the semiconductor chip **150**, and at least a portion of the electromagnetic shield structure **130**. The sealing member **140** may protect the semiconductor chip **150** and the electromagnetic shield structure **130** from mechanical shock and/or contamination.

In some embodiments, the sealing member **140** may include an epoxy-group resin or a polyimide-group resin. For example, the sealing member **140** may include an epoxy molding compound (EMC) and/or a photosensitive material such as a photo imagable encapsulant (PIE). In some embodiments, the sealing member **140** may include a base resin including an epoxy-group resin or a polyimide-group resin and a filler contained in the base resin **181**. (See, e.g., elements **181** and **183** of FIG. 9, wherein filler **183** may include an inorganic filler (e.g., silica) and/or an organic filler).

The sealing member **140** may include an outer portion **141** covering an outer surface of the electromagnetic shield structure **130**, an inner portion **142** provided between an inner surface of the electromagnetic shield structure **130** and the semiconductor chip **150**, first hole fillings **143** filling the first through holes **133** of the side cover **131**, and second hole fillings **144** filling the second through holes **137** of the upper cover **135**. The inner portion **142** of the sealing member **140** may cover the upper surface **155** and the sidewalls **157** of the semiconductor chip **150**, and may fill a gap between the semiconductor chip **150** and the package substrate **101**, thereby contacting the chip connecting bumps **171**. The outer portion **141**, the inner portion **142**, the first

hole fillings **143**, and the second hole fillings **144** of the sealing member **140** may include one or more of the same (or different) material(s).

In some embodiments, in order to form the sealing member **140**, constituent sealing material(s) (e.g., **180** of FIG. 9) of the sealing member **140** may be provided on the outer surface of the electromagnetic shield structure **130**. The sealing material(s) **180** provided on the outer surface of the electromagnetic shield structure **130** may be introduced to the inside of the electromagnetic shield structure **130** through the first through holes **133** of the side cover **131** and/or the second through holes **137** of the upper cover **135**. The sealing material(s) **180** introduced to the inside of the electromagnetic shield structure **130** may cover the upper surface **155** and the sidewalls **157** of the semiconductor chip **150**. Also, the sealing material(s) **180** introduced to the inside of the electromagnetic shield structure **130** may fill a gap between the semiconductor chip **150** and the package substrate **101**, as well as substantially surround the chip connecting bumps **171**.

In some embodiments, the side cover **131** may have a rectangular ring-like shape extending around the circumference of the semiconductor chip **150** having a rectangular shape. The side cover **131** having a rectangular ring-like shape may include four (4) segments **1311**, **1312**, **1313**, and **1314** linearly extending in relation to the first and second horizontal directions. For example, the side cover **131** may include a first segment **1311** and a second segment **1312** extending in the second horizontal direction, as well as a third segment **1313** and a fourth segment **1314** extending in the first horizontal direction. Here, the first through holes **133** may be formed at regular intervals in each of the four segments **1311**, **1312**, **1313**, and **1314** of the side cover **131**. Assuming this exemplary configuration, during the forming of the sealing member **140**, the sealing material **180** may be introduced towards the semiconductor chip **150** from different directions through the first through holes **133** penetrating the four segments **1311**, **1312**, **1313**, and **1314** of the side cover **131**. In this manner, the internal space **139** of the electromagnetic shield structure **130** may be sufficiently filled by the sealing member **140**.

In some embodiments, the first size **133D** of the first through hole **133** and the second size **137D** of the second through holes **137** may each be greater than a particle size (e.g., element **183D** of FIG. 9) of the filler **183** included in the sealing member **140** in order to readily facilitate the introduction of the sealing material **180** through the first through holes **133** and the second through holes **137** of the electromagnetic shield structure **130**. In some embodiments, the first size **133D** of the first through hole **133** and the second size **137D** of the second through holes **137** may be at least twice the particle size **183D** of the filler **183** included in the sealing member **140**. In some embodiments, when the diameter of the filler **183** included in the sealing member **140** is about 55 micrometers (μm), the first size **133D** of the first through hole **133** and the second size **137D** of the second through holes **137** may be at least about 110 μm .

In some embodiments, the electromagnetic shield structure **130** may be configured to exhibit a shielding effectiveness (SE) of at least 99% for a defined frequency band. Here, the SE may be defined by Equation (1) below, wherein ‘ λ ’ denotes a wavelength of electromagnetic energy incident to the electromagnetic shield structure **130**, ‘ l ’ denotes a maximum width of through holes provided in the electromagnetic shield structure **130** (e.g., a first maximum width for the first through holes **133** and a second maximum width for the

second through holes **137**), and ‘n’ denotes a number of through holes provided in the electromagnetic shield structure **130**.

$$S.E.[dB] = 20\log\left(\frac{\lambda}{2l}\right) - 10\log n \quad \text{Equation (1)}$$

In this regard, Table 1 below is a conversion table in which the SE per decibel (dB) is converted into the SE per percentage (%):

TABLE 1

Shielding Effectiveness (dB)	Shielding Efficiency (%)
0	0
10	90
20	99
30	99.39
40	99.99
50	99.999
60	99.9999
70	99.99999
80	99.999999
90	99.9999999

In some embodiments, the electromagnetic shield structure **130** may be configured to exhibit an SE of at least 99% for a defined frequency band based on Equation (1) and Table 1. In some embodiments, the first size **133D** of the first through holes **133** and the second size **137D** of the second through holes **137** provided in the electromagnetic shield structure **130** and/or the numbers of the first through holes **133** and the second through holes **137** provided in the electromagnetic shield structure **130** may be variously adjusted in order to ensure that the electromagnetic shield structure **130** exhibits an SE of at least 99% for the defined wavelength band.

In some embodiments, when the electromagnetic shield structure **130** is configured to shield electromagnetic waves having a first wavelength λ_1 , the first size **133D** of the first through hole **133** and the second size **137D** of the second through holes **137** may each be $\lambda_1/20$ or less or $\lambda_1/50$ or less. In this regard, if the first size **133D** of the first through hole **133** and/or the second size **137D** of the second through holes **137** are too large, the SE associated with the first wavelength λ_1 may rapidly deteriorate.

In some embodiments assuming that the semiconductor chip **150** is an RFIC configured to radiate electromagnetic energy at the first wavelength λ_1 (e.g., a maximum operation frequency), the first size **133D** of the first through holes **133** and the second size **137D** of the second through holes **137** may each be set to $\lambda_1/50$ or less. For example, further assuming that the maximum operation frequency of the semiconductor chip **150** is 11.660 Gbps (5.830 GHz), the first size **133D** of the first through holes **133** and the second size **137D** of the second through holes **137** may each be set to 1.03 mm or less.

Alternately, a degree to which electromagnetic energy is shielded by the electromagnetic shield structure **130** may be expressed in terms of skin depth. Here, “skin depth” refers to a propagation length of electromagnetic energy until the intensity of the electromagnetic energy absorbed by the electromagnetic shield structure **130** is attenuated to $1/e$. In this regard, skin depth may be represented by Equation (2) below, wherein ‘ δ ’ denotes skin depth, ‘ ρ ’ denotes resistivity

($\mu\Omega$ cm) of the electromagnetic shield structure **130**, ‘f’ denotes the frequency of the electromagnetic energy, and ‘ μ ’ denotes permeability of the electromagnetic shield structure **130**.

$$\text{skin depth } \delta = \sqrt{\frac{\rho}{\pi \times f \times \mu}} \quad \text{Equation (2)}$$

For example, assuming electromagnetic energy in a high frequency band, the skin depth may be relatively short and the length of the electromagnetic energy traveling within the electromagnetic shield structure **130** is relatively small. Therefore, the high-frequency electromagnetic energy may be adequately shielded by the electromagnetic shield structure **130** having a relatively thin thickness. In contrast, assuming electromagnetic energy in a low frequency band, the skin depth may be relatively long and the length of the electromagnetic energy traveling within the electromagnetic shield structure **130** is relatively long. Therefore, the thickness of the electromagnetic shield structure **130** required to adequately shield the electromagnetic energy in a low frequency band must be relatively thicker.

FIG. 4 is a graph illustrating a relationship between a thickness of the electromagnetic shield structure **130** and a minimum shielding frequency that may be adequately shielded by the electromagnetic shield structure **130**.

In FIG. 4, the horizontal axis represents the thickness of the electromagnetic shield structure **130**, and the vertical axis represents the minimum shielding frequency (i.e., the minimum frequency of electromagnetic waves that may be shielded by the electromagnetic shield structure **130**).

Referring to FIGS. 1, 2, 3, and 4, the minimum shielding frequency may be adjusted by adjusting the thickness of the electromagnetic shield structure **130**. That is, by adjusting a first thickness **131T** of the side cover **131** and a second thickness **135T** of the upper cover **135**, the range of shielding frequencies shielded by the electromagnetic shield structure **130** may be adjusted. In some embodiments, the first thickness **131T** of the side cover **131** and the second thickness **135T** of the upper cover **135** may be adjusted across a range of from about 200 μm to about 1000 μm . Accordingly, as indicated in the graph of FIG. 4, a minimum frequency of electromagnetic energy shielded by the electromagnetic shield structure **130** may range from about 0.004 MHz to about 0.11 MHz. Thus, the electromagnetic shield structure **130** may be used to effectively shield electromagnetic energy in a low frequency band.

FIG. 5 is a cross-sectional view illustrating a semiconductor package **1001** according to embodiments of the inventive concept. Hereinafter, the semiconductor package **1001** of FIG. 5 will be described primarily in relation to differences from the semiconductor package **1000** of FIGS. 1, 2 and 3.

Referring to FIG. 5, in addition to the package substrate **101**, the semiconductor chip **150** and the sealing member **140**, the semiconductor package **1001** may include an electromagnetic shield structure **130a** having a structure including multiple, stacked material layers.

In some embodiments, a side cover **131a** of the electromagnetic shield structure **130a** includes a first side shielding layer **191** of first material type, and a second side shielding layer **192** of second material type, different from the first material type. The second side shielding layer **192** of the side cover **131a** may be stacked on the first side shielding layer **191** of the side cover **131a**. For example, the first side

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shielding layer 191 of the side cover 131a may face the sidewalls 157 of the semiconductor chip 150, and the second side shielding layer 192 of the side cover 131a may be spaced apart from the sidewalls 157 of the semiconductor chip 150 by the first side shielding layer 191 of the side cover 131a interposed therebetween. In some embodiments, one of the first material type and the second material type may include Cu, and the other one of the first material type and the second material type may include a magnetic material (e.g., ferrite).

In some embodiments, an upper cover 135a may include a first upper shielding layer 193 of third material type and a second upper shielding layer 194 of fourth material type, different from the third material type. The second upper shielding layer 194 of the upper cover 135a may be stacked on the first upper shielding layer 193 of the upper cover 135a. For example, the first upper shielding layer 193 of the upper cover 135a may face the upper surface 155 of the semiconductor chip 150, and the second upper shielding layer 194 of the upper cover 135a may be spaced apart from the upper surface 155 of the semiconductor chip 150 by the first upper shielding layer 193 of the upper cover 135a interposed therebetween. The first upper shielding layer 193 of the upper cover 135a may be connected to the first side shielding layer 191 of the side cover 131a, and the second upper shielding layer 194 of the upper cover 135a may be connected to the second side shielding layer 192 of the side cover 131a. In some embodiments, the third material type may be the same as the first material type, and the fourth material type may be the same as the second material type. Alternately, in some embodiments, the third material type may be different from the first material type, and the fourth material type may be different from the second material type.

Although the electromagnetic shield structure 130a of FIG. 5 has been described in relation to a two layer structure, those skilled in the art will appreciate that more than two layers may be used to form the electromagnetic shield structure 130a. For example, the side cover 131a and the upper cover 135a may each have a structure in which three or more material layers are stacked.

FIG. 6 is a cross-sectional view illustrating a semiconductor package 1002 according to embodiments of the inventive concept. Hereinafter, the semiconductor package 1002 of FIG. 6 will be described primarily in relation to differences from the semiconductor package 1000 of FIGS. 1, 2 and 3.

Referring to FIG. 6, the semiconductor package 1002 may additionally include an underfill material layer 175.

Here, the underfill material layer 175 may be provided between the semiconductor chip 150 and the package substrate 101 to substantially surround the chip connecting bumps 171. The underfill material layer 175 may include an epoxy resin and may be formed using a capillary underfill method. In some embodiments, the underfill material layer 175 may be formed using a non-conductive film. During a process of manufacturing the semiconductor package 1002, after formation of the underfill material layer 175 is complete, mounting of the electromagnetic shield structure 130 and formation of the sealing member 140 may be sequentially performed. The sealing member 140 may cover the upper surface 155 and the sidewalls 157 of the semiconductor chip 150 and may cover side surfaces of the underfill material layer 175.

FIG. 7 is a cross-sectional view illustrating a semiconductor package 1003 according to embodiments of the inventive concept. Hereinafter, the semiconductor package

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1003 of FIG. 7 will be described primarily in relation to differences from the semiconductor package 1000 of FIGS. 1, 2 and 3.

Referring to FIG. 7, the semiconductor package 1003 may further include an antenna pattern 179.

Here, the antenna pattern 179 may be configured to communicate RF signals. The antenna pattern 179 may be electrically connected to the semiconductor chip 150 including an RFIC and may transmit an RF signal in response to operation of the semiconductor chip 150 and/or receive an RF signal from an outside source. The antenna pattern 179 may be disposed on the package substrate 101. The antenna pattern 179 may include a metal. The antenna pattern 179 may have a structure suitable for the communication of RF signals. For example, the antenna pattern 179 may include a patch antenna, a dipole antenna, a meander, etc.

The antenna pattern 179 may be provided on the upper surface of the package substrate 101 and may be adjacent to the side cover 131 of the electromagnetic shield structure 130. In this case, an RF signal communicated by the antenna pattern 179 may be transmitted through a through hole 134 of the side cover 131 adjacent to the antenna pattern 179. In this regard, the through hole 134 of the side cover 131 adjacent to the antenna pattern 179 may have a size suitable for the communication of the RF signal and may have a size greater than that of the first through holes 133 provided in the side cover 131.

FIGS. 8A to 8E are related cross-sectional views illustrating in one example a method of manufacture for the semiconductor package 1000 of FIGS. 1, 2 and 3 according to embodiments of the inventive concept, and FIG. 9 is a cross-sectional view of further illustrating the forming of the sealing member 140.

Referring to FIG. 8A, the semiconductor chip 150 may be mounted on a package substrate 101a. The package substrate 101a may have a planar area in which two or more semiconductor chips 150 may be mounted. The semiconductor chip 150 may be mounted on the package substrate 101a in a face-down (or flip-chip) manner. The chip pads 153 of the semiconductor chip 150 may be electrically and physically connected to the first upper substrate pads 121 of the package substrate 101a through the chip connecting bump 171.

Referring to FIG. 8B, the electromagnetic shield structure 130 is mounted on the package substrate 101a. The electromagnetic shield structure 130 may be mounted on the package substrate 101a to cover the semiconductor chip 150. The side cover 131 of the electromagnetic shield structure 130 may be electrically and physically connected to the second upper substrate pads 122 of the package substrate 101a through the conductive bonding material layer 173.

Referring to FIG. 8C, the sealing member 140 configured to cover and seal the electromagnetic shield structure 130 and the semiconductor chip 150 is formed on the package substrate 101a. The sealing member 140 may cover the outer surface of the electromagnetic shield structure 130, fill the first through holes 133 and the second through holes 137 of the electromagnetic shield structure 130, and fill the internal space (139 of FIG. 9) of the electromagnetic shield structure 130. In some embodiments, the sealing member 140 may fill a gap between the semiconductor chip 150 and the package substrate 101a and may contact the chip connecting bumps 171.

Referring to FIGS. 8C and 9, to form the sealing member 140, the sealing material 180 in a liquid state may be supplied onto the outer surface of the electromagnetic shield

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structure 130. The sealing material 180 provided on the outer surface of the electromagnetic shield structure 130 may be introduced to the inside of the electromagnetic shield structure 130 through the first through holes 133 of the side cover 131 and the second through holes 137 of the upper cover 135. The sealing material 180 introduced to the inside of the electromagnetic shield structure 130 fills the internal space 139 of the electromagnetic shield structure 130 and may cover the upper surface 155 and the sidewalls 157 of the semiconductor chip 150. Also, the sealing material 180 introduced to the inside of the electromagnetic shield structure 130 may fill a gap between the semiconductor chip 150 and the package substrate 101a and surround the chip connecting bumps 171. The first size 133D of the first through holes 133 and the second size 137D of the second through holes 137 may be greater than the particle size 183D of the filler 183 included in the sealing material 180, such that the sealing material 180 may be smoothly introduced to the inside of the electromagnetic shield structure 130 through the first through holes 133 of the side cover 131 and the second through holes 137 of the upper cover 135. For example, the first size 133D of the first through holes 133 and the second size 137D of the second through holes 137 may be at least twice that of the particle size 183D of the filler 183 included in the sealing material 180. Thereafter, the sealing member 140 may be formed by curing the sealing material 180.

Referring to FIG. 8D, the first external connection terminals 161 and the second external connection terminals 162 are attached onto the first lower substrate pads 123 and the second lower substrate pads 124 of the package substrate 101a. For example, to form the first external connection terminals 161 and the second external connection terminal 162, solder balls may be attached to the first lower substrate pads 123 and the second lower substrate pads 124 and a reflow process may be performed on the solder balls.

Referring to FIG. 8E, the resulting structure of FIG. 8D may be divided (or singulated) into a number of semiconductor packages 1000, each having a single package size by (e.g.) performing a sawing process. That is, by physically cutting and separating the resulting structures of FIG. 8D along scribe lane SL, individual semiconductor packages 1000 may be obtained.

According to embodiments of the inventive concept, since the semiconductor package 1000 includes an effectively designed electromagnetic shield structure 130 configured to properly shield electromagnetic energy, degradation in performance of the semiconductor package 1000 due to electromagnetic interference may be prevented.

Also, according to embodiments of the inventive concept, since the semiconductor package 1000 includes the sealing member 140 configured to seal the semiconductor chip 150 and the electromagnetic shield structure 130, potential damage to and/or contamination of the semiconductor chip 150 and the electromagnetic shield structure 130 may be prevented. Also, since the electromagnetic shield structure 130 includes through holes configured to allow introduction of a sealing material, the sealing member 140 for sealing the semiconductor chip 150 and the electromagnetic shield structure 130 may be formed through a single sealing process.

While the inventive concept has been particularly shown and described with reference to certain illustrated embodiments thereof, it will be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the inventive concept, as defined by the following claims.

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What is claimed is:

1. A semiconductor package comprising:

a package substrate including a substrate pad, the substrate pad configured to be electrically grounded;
a semiconductor chip on the package substrate;
an electromagnetic shield structure on the package substrate and including an upper cover covering an upper surface of the semiconductor chip and a side cover surrounding the semiconductor chip; and
a sealing member comprising an insulating material contacting the semiconductor chip and the electromagnetic shield structure,
wherein the side cover has first through holes extending laterally through the side cover,
wherein the upper cover has second through holes extending vertically through the upper cover,
wherein the electromagnetic shield structure is electrically connected the substrate pad, and
wherein the sealing member extends into the first through holes and the second through holes.

2. The semiconductor package of claim 1, wherein the sealing member includes an outer portion covering outer surfaces of the electromagnetic shield structure, an inner portion between inner surfaces of the electromagnetic shield structure and the semiconductor chip, first hole fillings filling the first through holes, and second hole fillings filling the second through holes.

3. The semiconductor package of claim 2, wherein the inner portion, the outer portion, the first hole fillings, and the second hole fillings include a same material.

4. The semiconductor package of claim 2, wherein the sealing member includes an epoxy molding compound.

5. The semiconductor package of claim 1, wherein a thickness of the upper cover and a thickness of the side cover range between about 200 μm to about 1,000 μm .

6. The semiconductor package of claim 1, wherein the sealing member includes a filler having a particle size, and a first size of the first through holes and a second size of the second through holes are at least twice the particle size of the filler.

7. The semiconductor package of claim 1, wherein the semiconductor chip is configured to radiate electromagnetic energy at a first wavelength λ of a maximum operation frequency, and a first size of the first through holes and a second size of the second through holes are each less than or equal to $\lambda/50$.

8. The semiconductor package of claim 1, wherein a first size of the first through holes and a second size of the second through holes each range between about 0.11 mm to about 1.03 mm.

9. The semiconductor package of claim 1, further comprising:

first upper substrate pads connecting the semiconductor chip to an internal interconnection pattern within the package substrate; and

wherein the substrate pad includes a second upper substrate pad configured to physically connect the electromagnetic shield structure.

10. The semiconductor package of claim 9, further comprising:

a conductive bonding material layer between the second upper substrate pad and the electromagnetic shield structure.

11. The semiconductor package of claim 9, further comprising:

chip connecting bumps between the first upper substrate pads and the semiconductor chip, wherein the sealing

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member fills a gap between the semiconductor chip and the package substrate and surrounds the chip connecting bumps.

12. The semiconductor package of claim 1, further comprising:

chip connecting bumps between first upper substrate pads on an upper surface of the package substrate and the semiconductor chip; and

an underfill material layer filling a gap between the semiconductor chip and the upper surface of the package substrate and surrounding the chip connecting bumps.

13. The semiconductor package of claim 1, wherein the side cover includes a first side shielding layer of a first material type and a second side shielding layer of a second material type different from the first material type, the second side shielding layer being stacked on the first side shielding layer, and

the upper cover includes a first upper shielding layer of a third material type and a second upper shielding layer of a fourth material type different from the third material type, the second upper shielding layer being stacked on the first upper shielding layer.

14. The semiconductor package of claim 1, further comprising:

an antenna pattern between the semiconductor chip and the side cover, electrically connected to the semiconductor chip, and configured to communicate radio frequency signals.

15. The semiconductor package of claim 1, wherein the side cover has a rectangular ring-like shape including four segments, and the first through holes penetrate each of the four segments.

16. A semiconductor package comprising:

a package substrate including a base, first upper substrate pads on an upper surface of the package substrate, a second upper substrate pad on the upper surface of the package substrate, and an internal interconnection pattern within the base and connected to the first upper substrate pads;

a semiconductor chip including chip connecting bumps respectively connected to the first upper substrate pads;

an electromagnetic shield structure on the package substrate and including an upper cover covering an upper surface of the semiconductor chip and a side cover surrounding the semiconductor chip, wherein the side cover is electrically connected to the second upper substrate pad and ground; and

a sealing member comprising an insulating material between the semiconductor chip and the electromagnetic shield structure,

wherein the side cover has first through holes extending in a first direction through the side cover,

the upper cover has second through holes extending in a second direction through the upper cover,

the second direction is different from the first direction, and

the sealing member extends into the first through holes and the second through holes.

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17. The semiconductor package of claim 16, wherein the sealing member includes a filler having a particle size, and a first size of the first through holes is at least twice that of the particle size,

wherein the first direction is a horizontal direction, and wherein the second direction is a vertical direction.

18. A semiconductor package comprising:

a package substrate including first upper substrate pads on an upper surface of the package substrate and a second upper substrate pad connected to ground and on the upper surface of the package substrate;

a radio frequency integrated circuit mounted on the package substrate using chip connecting bumps respectively connected to the first upper substrate pads, wherein the radio frequency integrated circuit is configured to communicate a radio frequency signal having a wavelength λ ;

an electromagnetic shield structure mounted on the package substrate and including an upper cover covering an upper surface of the radio frequency integrated circuit and a side cover surrounding side surfaces of the radio frequency integrated circuit, wherein the side cover is penetrated by regularly spaced first through holes and the upper cover is penetrated by regularly spaced second through holes; and

a sealing member comprising an insulating material disposed between the electromagnetic shield structure and the radio frequency integrated circuit, surrounding the chip connecting bumps, and including a filler having a particle size and filling the first through holes and the second through holes,

wherein a thickness of the upper cover and a thickness of the side cover range between about 200 μm to about 1,000 μm , a first size of the first through holes and a second size of the second through holes are each at least twice that of the particle size, and the first size of the first through holes and the second size of the second through holes are each less than or equal to $\lambda/50$,

wherein the electromagnetic shield structure is electrically connected to the second upper substrate pad,

wherein the side cover has first through holes extending laterally through the side cover, and

wherein the upper cover has second through holes extending vertically through the upper cover.

19. The semiconductor package of claim 18, wherein the thickness of the upper cover and the thickness of the side cover are the same,

the first size of the first through holes and the second size of the second through holes are the same, and

a first distance between centers of two adjacent first through holes and a second distance between centers of two adjacent second through holes are the same.

20. The semiconductor package of claim 18, further comprising:

a conductive bonding material layer between the second upper substrate pad and the side cover of the electromagnetic shield structure.

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